Chromatic Accessibility in Latin American University Websites: A Large-Scale Survey of WCAG Contrast Compliance

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Abstract

Web accessibility audits at scale have historically concentrated on North American and European institutions, leaving higher education in Latin America largely unexamined. We crawled the homepages of 464 Latin American university websites across 17 countries and successfully extracted computed text/background colour pairs from 259 of them (41,360 pairs in total). Each pair was converted from sRGB to CIELAB and evaluated against the WCAG 2.2 contrast thresholds, with the axe-core accessibility engine as an automated cross-check ($r = 0.347$). We find that 21.6% of all colour pairs fail the AA threshold, and 93.4% of sites contain at least one critical failure. A pair-level chi-square test suggests compliance differs by country ($\chi^2 = 1625.04$, $df = 16$, $p < 0.001$, Cramér's $V = 0.198$), but because colour pairs are clustered within sites and not independent, we treat this as exploratory: a site-level test on the true unit of analysis (255 sites, countries with $\geq$5 sites) is not significant ($\chi^2 = 22.92$, $df = 15$, $p = 0.086$). Larger text fails slightly more often than body text, and only continuous font size keeps a small association with failure under site-clustered standard errors ($p = 0.027$). Clustering the failing pairs' text colours in CIELAB shows that the largest group (60.1%) is white text resolved against a white or near-white background (median ratio 1.01), most plausibly text drawn over background images that colour extraction cannot see. Mid-grey text on white, the pattern usually assumed, accounts for 19.9%, and saturated colours for the remaining 20.0%.

Keywords: Colour contrast, Chromatic accessibility, WCAG 2.2, CIELAB colour space, Web accessibility, Information design.

1 Introduction

Web accessibility is frequently framed as a compliance checkbox rather than as a problem of information design: which visual choices allow a message to be perceived by the range of readers who will actually encounter it. Colour contrast between text and its background is one of the most consequential of those choices, because unlike many other accessibility barriers it is silent: a page can look complete to a sighted designer working under ideal lighting on a calibrated monitor while remaining functionally illegible to a large fraction of visitors.

Large-scale, empirical audits of contrast compliance exist, most notably WebAIM's annual scan of the top one million global home pages (WebAIM, 2026), which finds low-contrast text on 83.9% of pages and identifies it as the most commonly detected accessibility issue, a position it has held in every edition since 2019. Such global scans report an aggregate world figure, though, and are not designed to, and do not, break results out for Latin American higher education specifically. Regional surveys of Latin American university websites exist (Acosta-Vargas, Acosta & Lujan-Mora, 2018), but they evaluate WCAG conformance as a whole; to our knowledge none has measured colour contrast specifically, from computed styles, at regional scale. The gap matters because university websites are a primary public-facing channel for admissions, financial aid, and academic information, and design conventions, content

management systems, and institutional resourcing in the region differ enough from those studied elsewhere that findings cannot simply be assumed to transfer.

This paper reports a large-scale survey of chromatic accessibility on the homepages of Latin American universities. Rather than relying on a small hand-picked sample, we build a stratified list of institutions from an open, community-maintained directory, crawl each homepage with a real browser engine to capture computed (post-CSS) colour pairs, and classify every pair against the WCAG 2.2 (World Wide Web Consortium (W3C), 2023) contrast thresholds. We ask two questions. First, at what rate do these institutions fail contrast requirements, and does that rate vary systematically by country? Second, what visual pattern, if any, characterises the colours involved in a failure? The second question matters because the answer determines what kind of intervention would actually help: a narrow pattern (e.g., light grey body text) suggests a simple style-guide fix, while a diffuse one suggests a more structural design-literacy problem.

The contributions of this work are: (1) a regional survey focused specifically on colour contrast in Latin American university homepages, measured from computed (post-CSS) styles on a stratified sample of 464 institutions across 17 countries; (2) a site-level, clustering-aware statistical treatment of the country question that distinguishes a descriptive pair-level pattern from a confirmed site-level effect; (3) an unsupervised characterisation, in the full CIELAB colour space, of the visual patterns underlying contrast failures in this region, showing that the most frequent failure signature is white text resolved against a light background, most plausibly an extraction artefact of text over images, while mid-grey text on white and saturated colours each account for about a fifth of failures.

2 Related Work

Contrast and legibility research inside the graphic design literature has largely focused on typography and reading performance, not on regulatory thresholds. Weingerl, Nedeljković & Pušnik (2022) measured how quickly five-letter words displayed on screen are recognised and found that recognition time depends on the typeface, the letter case, and the position of the word on the screen; legibility therefore depends on several presentation variables at once, which is why we analyse contrast together with font size and weight. Punsongserm & Suvakunta (2025) conducted a technical review of typeface choice, type size, and colour contrast in government mobile applications, and, closer to our own setting, identified legibility and colour-contrast problems in public-sector mobile applications and argued that existing government standards do not address them in sufficient detail; we examine the contrast dimension of that problem, in a different modality and region, for university websites. In a related display context, Jović et al. (2026) review the literature on text legibility in augmented reality and identify colour and contrast among the ten parameters that govern it, together with background influence and environmental illumination. Ofosu-Asare (2024) evaluated and redesigned an online learning platform with 150 participants using heuristic evaluation, guideline reviews, and cognitive walkthroughs; expert- and user-based evaluations of this kind are informative but do not scale to a whole region, which is the gap this paper addresses for the chromatic dimension with an automated survey. Within this journal, colour and ink have mostly been studied on the print side, for example the drying systems used for printing inks (Saad et al., 2021); the screen-side counterpart, whether the sRGB colours chosen for web interfaces are distinguishable enough to be read, is the problem this paper addresses.

The conceptual foundation of our study rests on the universal web accessibility argument advanced by Henry, Abou-Zahra & Brewer (2014), who keep accessibility centred on people with disabilities while relating it to universal design and to the situational limitations, such as device or environmental constraints, that affect all users. Insufficient contrast is a case where both concerns meet, since it also hinders reading on poor screens or in bright light, so WCAG contrast compliance is better seen as a baseline quality requirement than as a minority accommodation. Rello & Baeza-Yates (2013) showed with eye-tracking that font choice alone measurably changes reading performance for readers with dyslexia, a population whose needs are not captured by automated WCAG pass/fail audits; this supports looking beyond a binary threshold test, which is why we also analyse failures in continuous CIELAB space.

The closest direct precedent in the Latin American higher-education domain is the body of work by Acosta-Vargas and collaborators, who have conducted the most systematic prior surveys of university website accessibility in the region. Their 2018 study (Acosta-Vargas, Acosta & Lujan-Mora, 2018) evaluated 348 Latin American university websites against WCAG 2.0, finding frequent violations (most often missing alternative text for images); it is the closest benchmark against which our 93.4% per-site failure rate and our 21.6% pair-level failure rate should be positioned. A subsequent heuristic method for users with low vision (Acosta-Vargas, Salvador-Ullauri & Lujan-Mora, 2019), applied to 40 websites, 30

of them Latin American universities, argued that automatic tools alone do not give an adequate picture of accessibility barriers; this is one reason we do not stop at the WCAG pass/fail ratio and also characterise failing pairs in the full CIELAB $L^*a^*b^*$ space. Their 2020 dataset (Acosta-Vargas, González & Luján-Mora, 2020) makes the evaluation data for 368 pages publicly available, enabling direct comparison with automated-tool outputs; our study is methodologically complementary in that we use a different pipeline (computed-style extraction via Playwright instead of a WAVE rule-engine scan) and restrict the analysis to the chromatic dimension specifically, extracting every text/background colour pair in CIELAB coordinates instead of a binary pass/fail per element, which allows us to characterise the failure patterns in continuous colour space. More recent work by the same group confirms that the contrast failure pattern extends beyond the university sector: Acosta-Vargas et al. (2022) found contrast errors to be the dominant accessibility barrier in 50 top-ranked e-commerce websites (83.1% of failures classified as "perceivable" violations), and a complementary study of 50 popular Android applications available in Ecuador (Acosta-Vargas et al., 2021) found text contrast among the recurring barriers (9.2% of detected problems), placing our university-website result within a broader pattern of accessibility problems in Latin American digital products.

3 Methodology

3.1 Site selection

We built the candidate list from the *Hipo/university-domains-list* open dataset, filtered to the 19 countries of Latin America (Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, Venezuela), and drew a stratified random sample across countries (capped per country to avoid Brazil and Mexico dominating the list); because of the cap, regional aggregates are unweighted averages over sampled institutions, not estimates weighted by each country's number of universities. The final list contained 464 domains across 17 countries. Bolivia and Venezuela are absent: the source dataset lists them under their official names ("Bolivia, Plurinational State of"; "Venezuela, Bolivarian Republic of"), which our country-name filter did not match, so they were excluded at the sampling stage (see Limitations).

3.2 Crawling

Each homepage was visited with a headless Chromium browser (Playwright), identifying our crawler with a descriptive user agent and honoring `robots.txt` (fetched with the same identifying user agent, since fetching it with a generic client triggers 403 responses on several sites and would otherwise cause valid, crawlable sites to be excluded by mistake). Requests were rate-limited to one every two seconds. Of the 464 domains, 278 (59.9%) were successfully rendered; the remainder were unreachable, timed out, or were excluded by `robots.txt` (see Table 1). The share of sampled domains that yielded analysable pairs ranged from 36% (Mexico, Paraguay) to 91% (Guatemala), but did not differ significantly by country ($\chi^2 = 21.68$, $df = 16$, $p = 0.154$).

3.3 Colour pair extraction

For each of the 278 reachable sites we revisited the live page a second time and, via `getComputedStyle()`, extracted every visible text node's computed text colour together with its *effective* background colour, resolved by walking up the DOM ancestor chain until a non-transparent background is found, since a text element's immediate parent is frequently transparent. Elements with zero width/height, `display:none`, `visibility:hidden`, or zero opacity were discarded. Colours with partial transparency (`rgba`, 7.5% of pairs) were composited over white before computing contrast, and element-level opacity between 0 and 1 was not applied; both are approximations of the colour actually rendered (Limitations). This step requires a live browser, not the raw HTML saved during crawling, because a serialised DOM does not inline externally linked CSS; computing style from the saved markup alone would silently fall back to browser defaults. This yielded 41,360 colour pairs from 259 sites (19 sites returned no usable pairs, typically due to navigation during evaluation).

3.4 Contrast calculation and cross-validation

Each pair's WCAG relative luminance and contrast ratio were computed directly from sRGB, and each pair was independently converted from sRGB to CIELAB (International Commission on Illumination

(CIE), 2007) (D_{65} illuminant) for the clustering analysis; colour distances are reported as CIE76 ΔE, the Euclidean distance in the same space used for clustering. As a cross-check, we additionally ran the `axe-core` accessibility engine (the library underlying most commercial accessibility linters) against the same live page and recorded its colour-contrast violation and pass counts, from which we compute each site's axe-core violation rate.

3.5 Classification and aggregation

Following WCAG 2.2, each pair was classified as *fail*, *AA*, or *AAA* using thresholds that depend on text size: 3:1 (AA) / 4.5:1 (AAA) for large text (≥24px, or ≥18.66px bold), and 4.5:1 (AA) / 7:1 (AAA) otherwise. Results were aggregated per site (median ratio, % of pairs failing AA, and whether the site has a critical failure, defined as at least one failing pair).

3.6 Statistical analysis

We tested independence between country and AA compliance with a χ^2 test (continuity-corrected, restricted to countries having ≥ 30 pairs) and we report Cramér's V as the effect size. This test, like any test run on the 41,360-pair table, treats colour pairs as independent observations, but pairs from the same site share a common design system and are not independent draws, a form of pseudo-replication that inflates the effective sample size and can make a null or weak site-level effect appear significant at the pair level. We therefore repeat the country comparison at the site level, the unit at which country actually varies: a χ^2 test on each site's binary critical-failure status, and a Kruskal-Wallis test on each site's continuous %AA-failure, both restricted to the 16 countries with ≥5 sites (255 of 259 sites). We fit a logistic regression of pair-level failure on country, font size, and whether the text qualifies as "large," and repeat it with site-clustered sandwich standard errors (generalised estimating equations, independence working correlation, clustered by site) to check which effects, not only the country effect, survive proper clustering. We clustered the failing pairs' text-colour coordinates in the full, standardised CIELAB $L^*a^*b^*$ space with k-means, rather than the a^*b^* chroma plane alone, since WCAG failure is a function of luminance and a chroma-only clustering would place every achromatic colour at the origin regardless of lightness; we computed the silhouette score for $k \in \{2, \ldots, 20\}$ as a sensitivity check and fixed $k = 6$ for interpretability (Results; the curve is in the repository).

4 Results

Table 1: Crawl outcomes for the 464 sampled domains.

Outcome	Count
Successfully rendered	278
Unreachable / timed out / other error	167
Excluded by `robots.txt`	19

Across the 41,360 extracted colour pairs, 21.6% fail the applicable WCAG AA threshold, 16.6% meet AA but not AAA, and 61.8% meet AAA (Figure 1). At the site level, 93.4% of the 259 successfully analysed homepages contain at least one critical (failing) pair, and the median site has 20.1% of its pairs failing AA, so failures are not concentrated in a handful of outlier institutions. Because a single failing pair is enough to mark a site, we also report stricter thresholds: 71.0% of sites have more than 10% of their pairs failing, and 41.7% more than 25%. This per-page failure rate is higher than WebAIM's 83.9% figure for the global top one million home pages (WebAIM, 2026), though the two figures come from different detection pipelines applied to different populations and should be read as an order-of-magnitude reference rather than a controlled comparison (Limitations).

Table 2 gives descriptive statistics by country before turning to the significance question. The pair-level test suggests compliance varies by country ($\chi^2 = 1625.04$, $df = 16$, $p < 0.001$, Cramér's $V = 0.198$; Figure 2), and the pair-level logistic regression finds most country coefficients individually significant (e.g., Paraguay: $\beta = -2.23$, $p < 0.001$; Cuba: $\beta = +0.57$, $p < 0.001$). We do not take this at face value: because pairs are clustered within sites, the pair-level test's effective sample size (41,360) overstates independent information, and its p-values should be read as optimistic. The site-level test, restricted to the 255 sites in the 16 countries with ≥5 sites, is *not* significant either on critical-failure status ($\chi^2 = 22.92$, $df = 15$,

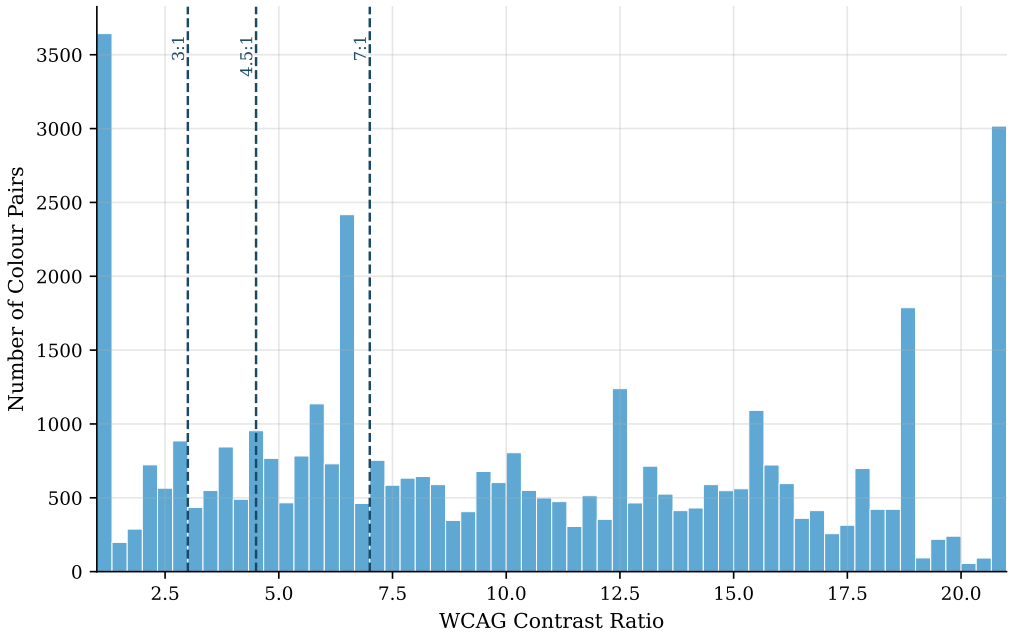


Figure 1: Distribution of WCAG contrast ratios across the 41,360 extracted colour pairs, with the thresholds marked; the 3:1 line applies only to large text, and the 4.5:1 and 7:1 lines are the AA and AAA thresholds for body text.

Table 2: Descriptive statistics by country (16 countries with ≥5 sites; 255 of 259 sites).

Country	Sites	Median % AA-fail	% sites, ≥1 critical fail
Argentina	27	22.0	96.3
Chile	27	15.0	96.3
Brazil	25	12.5	92.0
Colombia	26	19.0	100.0
Ecuador	24	20.1	91.7
Peru	21	25.4	81.0
Costa Rica	19	24.0	94.7
Mexico	15	9.0	73.3
Dominican Republic	15	33.6	100.0
El Salvador	13	15.5	100.0
Guatemala	10	22.3	100.0
Nicaragua	9	33.3	88.9
Panama	7	14.3	100.0
Cuba	6	31.6	100.0
Honduras	6	22.9	83.3
Paraguay	5	15.6	100.0

$p = 0.086$) or on continuous %AA-failure (Kruskal-Wallis $H = 20.55$, $p = 0.152$). Because the site-level critical-failure rate is high (most countries have few or no compliant sites), several cells of the country × failure contingency table have an expected count well below the conventional rule-of-thumb minimum of 5 for the chi-squared approximation to be reliable (minimum expected count = 0.33); as a sensitivity check we therefore also ran a Monte Carlo permutation test with 10,000 permutations, shuffling critical-failure status across sites while keeping each country's number of sites fixed and recomputing χ^2 each time. It agrees with the asymptotic result ($p_{\mathrm{MC}} = 0.0856$ versus $p = 0.0858$), so the result does not depend on the chi-squared approximation. We therefore report the country effect as a real but unresolved pattern (visible and directionally consistent at the pair level, but not confirmed at the site level with the sample sizes available here), not as a confirmed finding, and we do not treat the specific per-country coefficients above as reliable individually. Typography varies within a site, so it is less exposed to the between-

site clustering concern. In raw terms, large text fails slightly more often than body text (25.1% versus 21.0% of pairs), despite its lower 3:1 threshold. The two typography variables are strongly correlated (point-biserial $r = 0.66$), so in the joint model the large-text coefficient is negative only conditionally on pixel size ($\beta = -0.10$); with site-clustered sandwich standard errors (generalised estimating equations, independence working correlation) it is not significant ($p = 0.473$, versus $p = 0.022$ pooled), and entered alone it is positive and not significant either ($\beta = +0.10$, $p = 0.391$). Continuous font size keeps a small positive association with failure under clustering, both jointly ($\beta = +0.0108$, $p = 0.027$) and alone ($\beta = +0.0085$, $p = 0.035$): larger text fails somewhat more often, plausibly because headings and banner text are the elements most often placed over images.

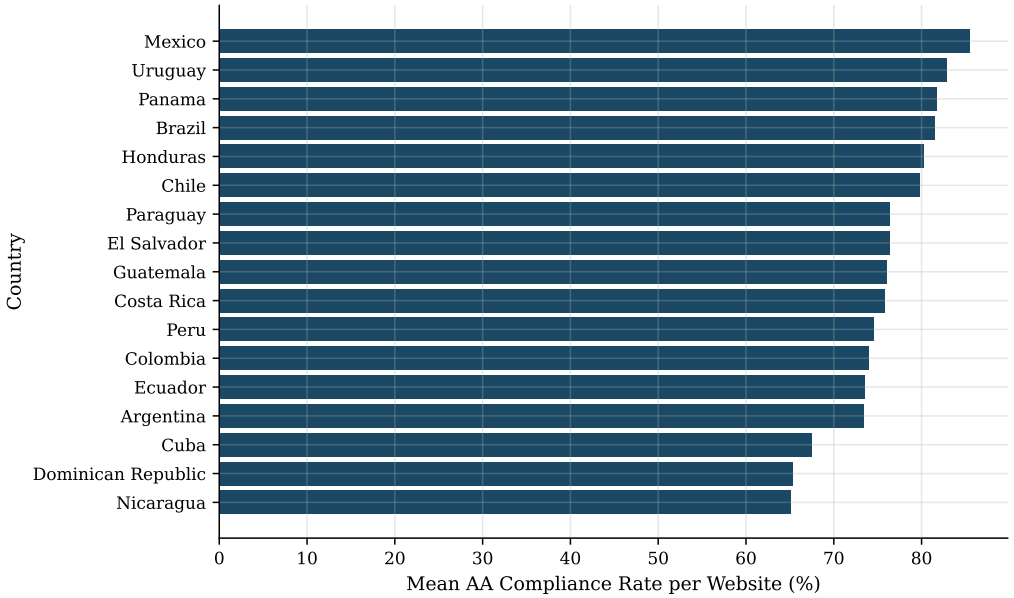


Figure 2: Mean AA compliance rate by country at the website level ($n \geq 3$ sites per country, 17 countries including Uruguay, 259 total sites; descriptive; see main text for why the associated significance test is not treated as confirmatory). Uruguay (4 sites) appears here but not in Table 2, which restricts descriptive statistics to the 16 countries with $n \geq 5$ sites (255 sites).

We clustered the 8,925 failing pairs' text-colour coordinates in the full, standardised CIELAB $L^*a^*b^*$ space with k-means at $k = 6$ (silhouette = 0.796); we cluster on all three coordinates, not a^*b^* alone, since WCAG failure is governed by luminance and a chroma-only projection would collapse every achromatic colour (black, grey, white) onto the origin regardless of its lightness, which would settle the composition question by construction. The silhouette score over $k \in \{2, \ldots, 20\}$ rises from 0.67 at $k = 2$ to about 0.80 at $k = 6$ and then stays between 0.77 and 0.83 without a clear peak, so we fix $k = 6$ for interpretability. The partition separates the low-chroma pairs into two clusters at different lightness levels. The larger is a near-white cluster (mean $L^* = 99.5$, chroma ≈ 0.2, 60.1% of failing pairs) in which 91.3% of the pairs have pure white text (#FFFFFF), the median background lightness is $L^* = 100$, and the median contrast ratio is 1.01: these are white texts resolved against white or near-white backgrounds, not grey text. The second is a mid-grey cluster (mean $L^* = 51.7$, chroma ≈ 5.0, 19.9%; median background $L^* = 99.2$), which does correspond to grey text on white. Four smaller chromatic clusters (4.1% to 6.1% of pairs each, chroma 41.6–78.4) span red, blue, yellow-green, and green regions of the plane (Figure 3). Across all failing pairs, 42.2% have a contrast ratio below 1.5, i.e. text that would be practically invisible if the recorded background were the rendered one, which points to the background-image artefact discussed below and in Limitations. Agreement between our per-site failure rate and `axe-core`'s per-site violation rate (violations over violations plus passes) is positive but moderate ($r = 0.347$, 253 sites with axe-core results), which is expected given the two tools differ in which elements they consider evaluable.

Table 3 shows that the single most frequent failing pair, by a wide margin, is white text on a white background (2,379 occurrences, $\Delta E = 0$). We cannot fully distinguish two explanations from the extracted data: text genuinely styled to match its background (a fully invisible pair), and text rendered over a photographic or gradient `background-image`, which our background-colour resolution (Methodology, Limitations) would also record as white-on-white, since it only inspects `background-color` up the

Table 3: The 10 most frequent failing colour pairs, with their WCAG contrast ratio and CIELAB colour difference ΔE (CIE76).

Text (hex)	Background (hex)	n pairs	Ratio	ΔE
#FFFFFF	#FFFFFF	2,379	1.00	0.0
#777777	#FFFFFF	186	4.48	50.0
#9F6B6B	#F2F2F2	172	3.92	50.2
#E33B3B	#FFFFFF	160	4.24	89.4
#FFFFFF	#EDEDED	137	1.17	6.3
#BFAC51	#FFFFFF	124	2.27	57.0
#999999	#FFFFFF	116	2.85	36.8
#00BCD4	#FFFFFF	87	2.30	48.8
#FFFFFF	#F2F2F2	81	1.12	4.5
#959595	#FFFFFF	76	3.00	38.3

DOM ancestor chain. Both explanations describe real accessibility problems, but only the first is literally invisible text. Together with the white-on-light-grey pairs in the same table, this signature makes up most of the near-white cluster.

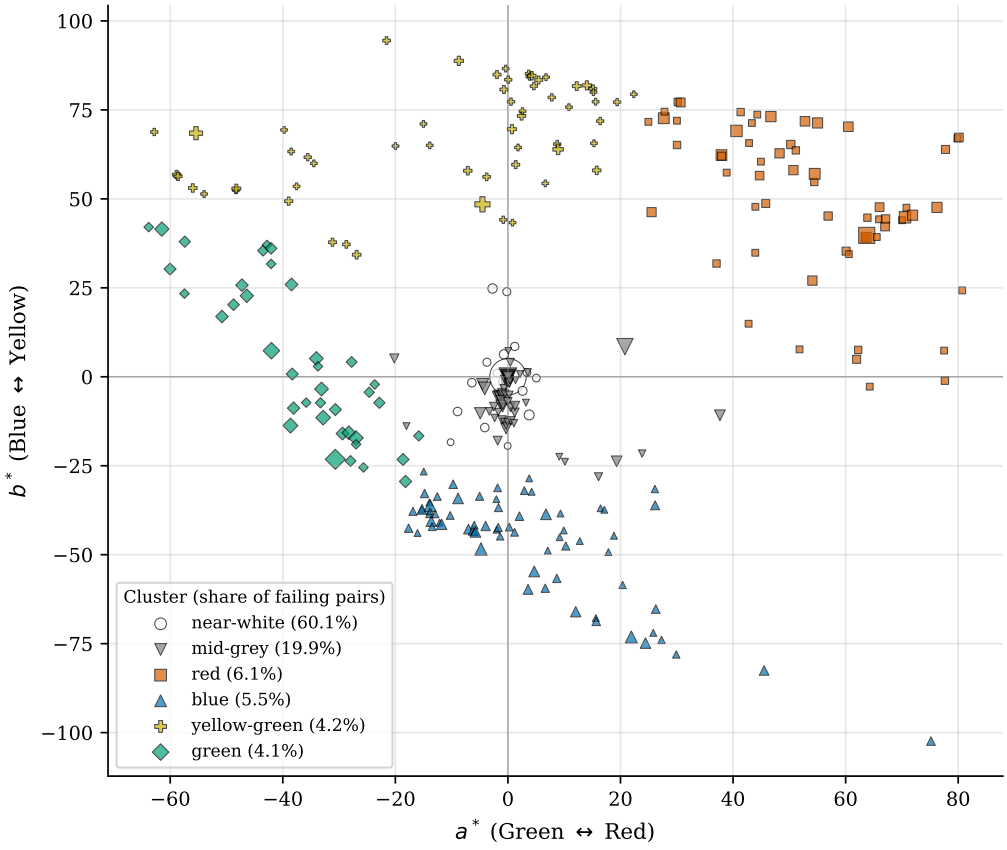


Figure 3: Failing pairs' text colours projected onto the CIELAB a^*b^* plane, one marker per distinct colour, with marker size proportional to the number of pairs and shape and shade identifying the k-means cluster ($k = 6$, computed in the full $L^*a^*b^*$ space). The near-white and mid-grey clusters overlap near the origin because they differ in lightness (L^*), which this projection does not show.

5 Discussion

That most Latin American university homepages fail WCAG contrast requirements mirrors findings from other regions and is, on its own, unsurprising. Our working hypothesis, drawn from common assumptions in the accessibility-audit literature, was that failures would concentrate in familiar patterns such as light grey text on white, text over unmasked photographic backgrounds, and unverified interaction states (hover/focus/visited). The clustering result supports that hypothesis only in part. Grey text on white, the pattern most often assumed, is real but accounts for about a fifth of failing pairs (the mid-grey cluster, 19.9%). The largest group (60.1%) is white text resolved against a white or near-white background, with a median contrast ratio of 1.01. That signature fits the second element of our hypothesis, text placed over photographs or gradients, because our extraction records the solid colour behind an image rather than the image itself; it can also include genuinely invisible text, and the data cannot separate the two. The remaining fifth consists of saturated colours used directly as text (4.1% to 6.1% each). The practical implication is therefore not a single style-guide fix: checking text placed over images and banners would address the largest group if the artefact reading is correct, a minimum-contrast rule for grey text would address the second, and checking brand colours used as text colours would address the third.

For the chromatic clusters, a plausible, though untested, explanation is institutional brand palettes (a university's primary colour used directly as a text colour, regardless of what it sits on), not a single stylistic trend (Figure 4 renders the most frequent grey and saturated failing pairs as measured). We have not tested this explanation against each institution's declared brand colours, and treat it as a hypothesis for future work rather than a finding.

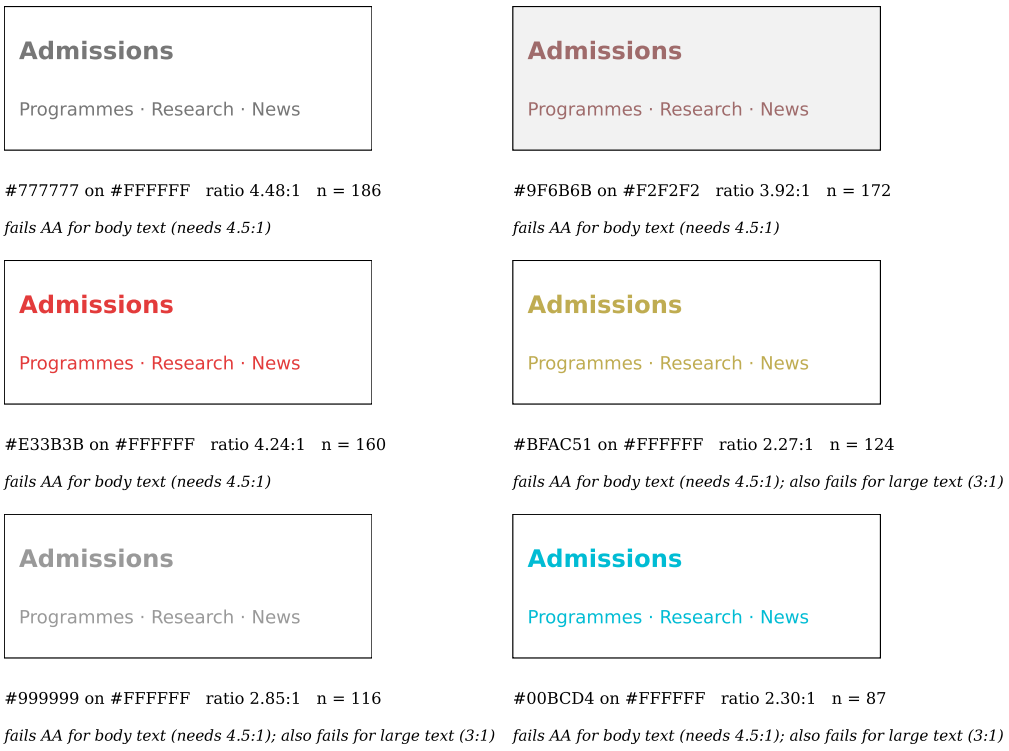


Figure 4: The six most frequent failing text/background pairs in Table 3 that contain visible text (white-on-white and white-on-light-grey pairs omitted), rendered with the measured colours. Each panel gives the colours, the WCAG contrast ratio, and the number of occurrences; the specimen text is illustrative and no institution is identified. Grey-on-white pairs and colours used directly as text (muted pink #9F6B6B on light grey, red, olive, cyan) both appear among the most frequent failures.

A note on the WCAG contrast formula itself, relevant to interpreting the residual variance our models leave unexplained: WCAG's ratio is based on relative luminance, not on a perceptually uniform distance measure. Two pairs with identical WCAG ratios can differ substantially in perceived contrast, which is part of why our CIELAB-space clustering was necessary in the first place: the WCAG ratio alone cannot distinguish the qualitatively different failure patterns we report.

6 Limitations

This study has several limitations. First, we crawled homepages only, not full sites; a homepage that passes may sit atop a site with far worse compliance on interior pages (or vice versa). Second, we measured only colour contrast, which is one of many WCAG success criteria; a chromatically compliant site can still be inaccessible on other dimensions (keyboard navigation, alt text, semantic structure). We also measured only the static rendered state of each page, so the interaction-state part of our initial hypothesis (hover, focus, and visited colours) was not tested. Third, the crawl captures a single point in time (August 2026); institutional websites are redesigned periodically, so these figures are not stable indefinitely. Fourth, Bolivia and Venezuela were excluded by a country-name mismatch in our sampling filter (the source lists them under their official names), so the 17-country sample does not cover every country in the region; we did not re-crawl them afterwards because a later crawl would not be comparable with the August 2026 snapshot. Fifth, and most consequential for how the country results should be read: colour pairs are clustered within sites (a single homepage contributes many pairs sharing one design system), so the pair-level χ^2 and logistic-regression tests overstate independent sample size and their p-values are optimistic; the site-level check we report alongside them (255 sites, $p = 0.086$) is not significant, so we treat the country pattern as suggestive, not confirmed, and do not rely on individual per-country coefficients. Sixth, 40.1% of the sampled domains could not be rendered or yielded no usable pairs; the rendering rate did not differ significantly by country (Results), but we had no institution-size or resourcing variable to test whether less-resourced institutions are under-represented, which would bias the country figures. Seventh, we used country as a proxy for institution type because a reliable public/private classification was not available across the source dataset; a finer-grained institutional variable, or a larger site-level sample powered to detect a modest effect, might resolve the country question more conclusively than either test here does. Eighth, our background-colour resolution walks the DOM ancestor chain for a non-transparent `background-color` and does not inspect `background-image`; a text element rendered over a photographic or gradient background resolves, incorrectly, against whatever solid colour sits further up the ancestor chain (typically page white). This is a plausible contributor to the single most frequent failing pair in our data (white text on a white background, 2,379 occurrences): at least part of that count is likely mislabelled text-over-image rather than genuinely invisible text, and we cannot currently distinguish the two cases from the extracted data alone. To bound how much this concern could be inflating our headline figures, we recomputed both under the most conservative reading, treating every single one of the 2,379 white-on-white pairs as an artefact and excluding them entirely: the pair-level failure rate drops from 21.6% to 16.8%, and the site-level share with at least one critical failure drops from 93.4% to 90.7% (only 7 of 259 sites, 2.7%, have white-on-white as their *only* recorded failure). Since some of these 2,379 pairs are probably invisible-text bugs and not background-image artefacts, this exclusion is an upper bound on the artefact's impact, not a corrected estimate; even so, both headline figures remain qualitatively unchanged. The same artefact affects the cluster analysis more strongly than the headline rates, because white-on-white pairs make up most of the largest cluster. Recording axe-core's *incomplete* results, which flag text over background images as not evaluable, would allow the artefact to be estimated rather than bounded; we did not store them in this crawl. Ninth, partially transparent colours were composited over white and partial element opacity was ignored, so the contrast of 7.5% of pairs is approximate. This also means our comparison to WebAIM's global figure (Results) uses a different detection instrument on a different population and should be read as an order-of-magnitude reference, not a controlled comparison. Finally, the moderate correlation with `axe-core` reflects methodological differences between contrast-ratio counting and axe-core's rule engine, not a validity problem with either approach, but it means the two should be read as complementary rather than interchangeable estimates.

7 Conclusion

A homepage-level survey of 259 Latin American university websites finds that WCAG AA contrast failures are widespread (93.4% of sites affected, 21.6% of all measured pairs); a country effect is visible at the pair level but is not confirmed once the actual unit of analysis (site) and the clustering of pairs within a site are accounted for, and we report it as an open question. The colours involved support our initial hypothesis only in part: in the full CIELAB $L^*a^*b^*$ space, grey text on white accounts for 19.9% of failing pairs and saturated colours for 20.0%, while the largest group (60.1%) is white text resolved against a white or near-white background, most plausibly text over images that colour extraction cannot see. Accessibility guidance for these institutions should therefore check text placed over images and banners, set a minimum contrast for grey text, and verify brand colours used as text colours; future audits should measure contrast

against the rendered background, not only the declared background colour.

Data Availability

The domain list, crawl logs, extracted colour pairs, and analysis code are permanently archived and openly accessible in Zenodo at `https://doi.org/10.5281/zenodo.22907769` and GitHub at `https://github.com/JuanAvilesP3/jged-contraste-cromatico`.

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Conflict of Interest

The authors declare that they have no conflict of interest.

Use of Generative AI

The authors used Claude (Anthropic), a generative AI large language model, as an assistant for language editing, support in writing and debugging analysis code, and LaTeX formatting. The research questions, study design, data collection, execution of the analyses, results, and their interpretation are the authors' own work; all AI-assisted output was reviewed and verified by the authors, who take full responsibility for the content of this article. No figure was produced with image-generation tools.

Author Contributions

Juan Pablo Aviles-Esparza: Conceptualization, Methodology, Software, Writing - Original Draft. **Italo Javier Tenempaguay-Granizo:** Data Curation, Formal Analysis, Validation, Visualization, Writing - Review & Editing. **Isaac David Torres-Paredes:** Supervision, Project Administration, Resources, Writing - Review & Editing.

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